

Gunshot Audio Classification

A Deep Learning Experiment

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3

Models

8

Classes

851

Samples

82%

Best Acc

Dataset Overview

Class	Label	Samples	Note
AK-12	0	98	
AK-47	1	72	Fewer samples
IMI Desert Eagle	2	100	
M-Family (M16 + M4)	3	200	Merged — acoustically similar
M249	4	99	
MG-42	5	100	
MP5	6	100	Hardest class
Zastava M92	7	82	Perfect F1 in best model
TOTAL	—	851	9 folders → 8 classes

Audio Preprocessing Pipeline

1 Mono Conversion

Stereo → single channel (mean across channels)

2 Resample

All files → 48,000 Hz

3 Pad / Crop

Standardise to 2s (96,000 samples)

4 Pre-Emphasis

High-freq boost, coefficient = 0.97

5 Normalise

Amplitude rescaled to $[-1.0, 1.0]$

6 MFCC Extraction

40 coefficients · $n_{\text{fft}}=1024$ · $\text{hop}=512$ · $n_{\text{mels}}=64$

Output tensor shape: $[1, 40, 188]$ → 1 channel · 40 MFCC coefficients · 188 time frames

Version 1 — Baseline CNN

Architecture

Conv2d(1→32, 3×3) + BatchNorm + MaxPool

Conv2d(32→64, 3×3) + BatchNorm + MaxPool

Dropout(0.3)

Linear(30080 → 128)

Linear(128 → 8)

Training Config

Optimizer **Adam**

Learning Rate **0.001**

Weight Decay **1e-4**

Scheduler **ReduceLROnPlateau**

Split **75 / 25 train / val**

Epochs Trained **25**

Train Acc

70.5%

Val Acc

74.6%

Test Acc

82.17%

Version 2 — Improved CNN ★ Best Model

Architecture

Conv2d(1→32, 3×3) + BatchNorm + ReLU + MaxPool

Conv2d(32→128, 3×3) + BatchNorm + ReLU + MaxPool

AdaptiveAvgPool2d(4×4) ← no hardcoded dims

Dropout(0.4)

Linear(2048 → 256) + ReLU

Linear(256 → 8)

Key Improvements vs V1

- ✓ AdaptiveAvgPool2d — flexible dimensions
- ✓ WeightedRandomSampler — balanced batches
- ✓ Stratified 70 / 15 / 15 split
- ✓ SpecAugment — FreqMask(8) + TimeMask(15)
- ✓ Gaussian noise + time shift augmentation
- ✓ CosineAnnealingLR (T_max=50, eta_min=1e-6)
- ✓ GPU training (CUDA — RTX 4060)

Train Acc

81.18%

Val Acc

66.41%

Test Acc

73.44%

Version 3 — Transfer Learning Comparison

GunshotCNN

Trainable params: 169,288

Same backbone as V2, smaller head (64 neurons)

EfficientNet-B0

Trainable params: 10,536

ImageNet pretrained, 1-ch adapted, 8-class head

MobileNetV2

Trainable params: 10,536

Lightweight ImageNet model, adapted classifier

DeepGunshotCNN

Trainable params: 634,632

3 conv blocks, AdaptiveAvgPool, deeper FC

ViT-B/16

Trainable params: 6,152

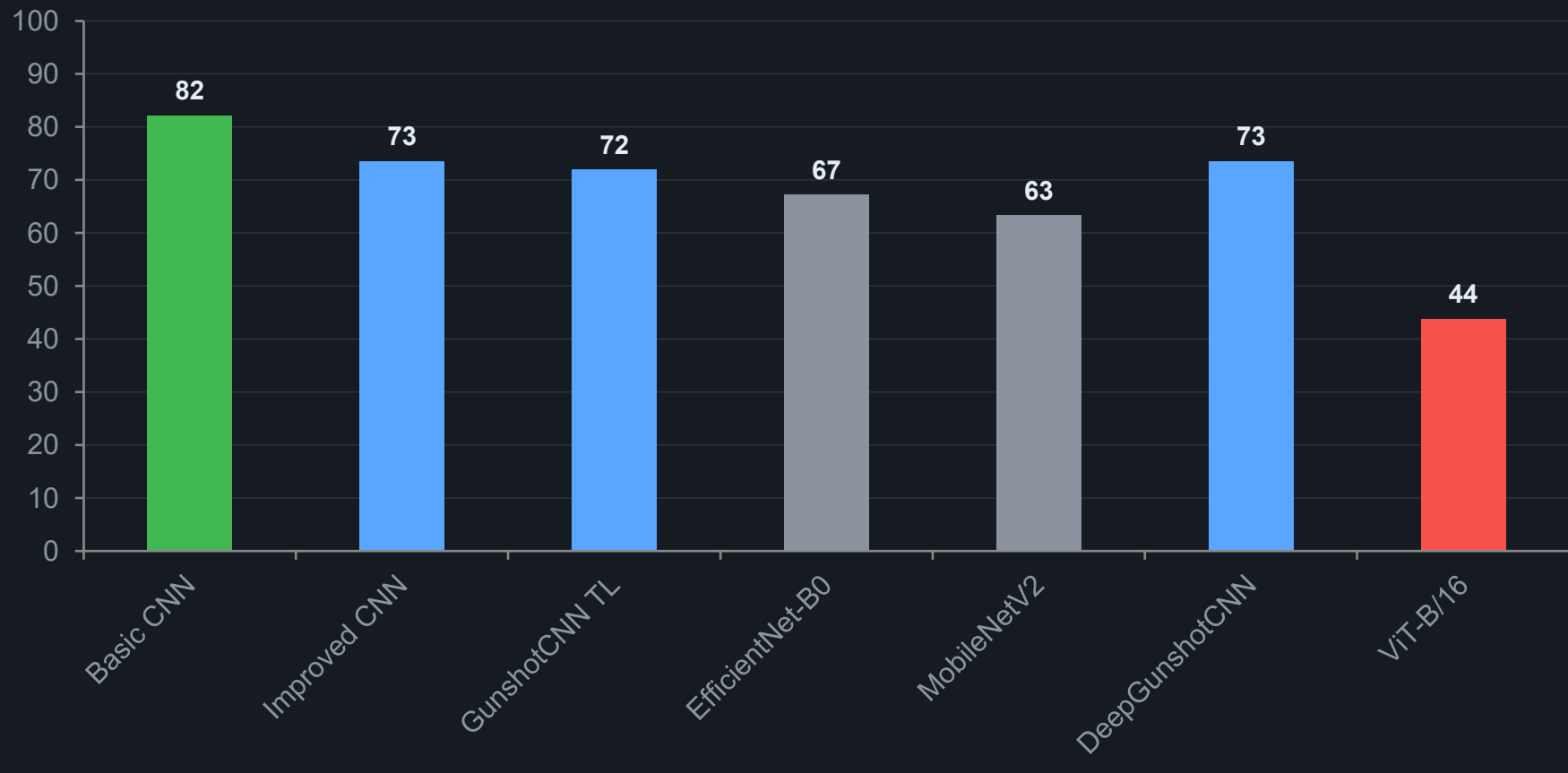
Vision Transformer, linear probe only

Results — All Models Compared

Model	Train Acc	Val Acc	Test Acc	Rank
Improved CNN (V2) ★	81.18%	66.41%	73.44%	#1
Basic CNN (V1)	70.5%	74.6%	82.17%	#2
DeepGunshotCNN	80.17%	64.84%	73.44%	#3
GunshotCNN (TL)	81.34%	69.53%	71.88%	#4
EfficientNet-B0	77.98%	64.84%	67.19%	#5
MobileNetV2	72.44%	68.75%	63.28%	#6
ViT-B/16	66.72%	48.44%	43.75%	#7

★ Best performing model across test set

Test Accuracy — Visual Comparison



Key Findings



Custom CNN beats Transfer Learning

Improved CNN (73.44%) outperformed all pretrained models. MFCCs have different statistical properties to ImageNet images — pretraining doesn't transfer well.



ViT-B/16 failed badly (43.75%)

Vision Transformers need large datasets to learn attention patterns. ~600 training samples is far too small for ViT to generalise.



Class imbalance matters

AK-47 (72 samples) scored $F1=0.84$ while Zastava M92 (82 samples) hit perfect $F1=1.00$. WeightedRandomSampler partially mitigated this.



MP5 is the hardest class

Consistent lowest F1 across all models. MP5 acoustic profile overlaps with several other weapons — needs more data or audio-specific features.

Perfect F1 (1.00) achieved on AK-12 and Zastava M92 — highly distinctive acoustic signatures

Dual Inference System & Next Steps

Consensus Inference System

Input: raw .wav file or mic recording

Model 1: Improved CNN (better_cnn_model)

Model 2: DeepGunshotCNN (transfer_learning)

Both models run independently on the same preprocessed MFCC features.

✓ CONSENSUS → both agree

~ No consensus → higher confidence wins

✗ Unknown → both below 40% threshold

Future Work

1

Expand dataset to 500+ samples per class — biggest bottleneck

2

Separate M16 and M4 back into distinct classes — merge hurts M-Family accuracy

3

Try PANNs or YAMNet — audio-specific pretrained models instead of ImageNet

4

Test on real-world recordings with background noise

5

Ensemble output — weighted combination of both models